Generative Models for Cat Image Synthesis

Deep Learning: Project III

Warsaw University of Technology

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Goal & Motivation

Train and compare three generative architectures on cat images at 64×64 resolution with a 6 GiB VRAM constraint.

Three approaches

1. **DCGAN** with Wasserstein loss and gradient penalty (WGAN-GP)
2. **VQ-VAE** with autoregressive PixelCNN prior
3. **DDPM** with DDIM fast sampling

Evaluation: Fréchet Inception Distance (FID) on 5 000 samples, visual inspection, and latent interpolation.

Real cat samples (64×64)

Dataset & Preprocessing

CAT Dataset (Kaggle)

- 9 997 RGB cat photos, varying resolutions
- Subdirectories CAT_00 to CAT_06

Preprocessing

1. Resize shorter edge to 64 px
2. CenterCrop 64×64
3. Normalize to $[-1, 1]$

Fixed seed 42. No augmentation in baselines. Augmentation tested in ablation: FID 284.05 ± 14.80, worse than baseline 264.25 ± 20.21.

DCGAN with WGAN-GP

Architecture

- Generator: $\mathbf{z} \in \mathbb{R}^{128}$, 4 transposed-conv blocks, Tanh
- Discriminator: 4 strided-conv (LeakyReLU), scalar logit

Training

- WGAN-GP: $\lambda_{GP} = 10$, $n_{\text{critic}} = 5$
- Adam $\beta_1 = 0.5$, $\text{lr} = 2 \times 10^{-4}$
- 100 epochs, batch 64

Note: WGAN-GP critic outputs a scalar score, not a probability — binary label targets have no role in this loss.

Why WGAN-GP? Smoother gradients, no weight-clipping issues. $n_{\text{critic}} = 5$ keeps the critic well-calibrated before each generator update.

Epoch 100

VQ-VAE with PixelCNN Prior

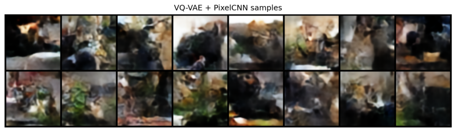
Phase 1: Autoencoder (100 epochs)

- 64×64 to 16×16 discrete code map
- Codebook: $K = 512$, $d = 64$, EMA updates ($\gamma = 0.99$)
- Commitment cost $\beta = 0.25$

Phase 2: PixelCNN Prior (100 epochs)

- 15-layer masked residual convolutions, 128 filters
- Trained on 256-token sequences (frozen encoder)
- Codebook perplexity: 403/512 (79% utilisation)

Why EMA codebook? Prevents codebook collapse. Gradient-based updates are unstable; early run with $K = 1024$ had only 30% code utilisation.



PixelCNN prior samples

DDPM with DDIM Sampling

Architecture

- U-Net, channel mults (1, 2, 2, 2), 2 res-blocks per level
- Self-attention at 16×16; EMA weights at inference

Diffusion process

- Linear schedule: $T = 1000$, $\beta_1 = 10^{-4}$, $\beta_T = 0.02$
- DDIM ($\eta = 0$): 50 steps gives $20\times$ speedup
- 200 epochs, batch 16 (6 GiB VRAM limit)

Why EMA decay 0.9999? Slower EMA reduces weight noise amplified by the long reverse chain. FID drops from 54.51 (no EMA) to 24.36 (with EMA) at 50 DDIM steps, a 55% improvement.

Epoch 200 (EMA weights)

FID Comparison

Fréchet Inception Distance (5 000 samples)

Model	FID ↓	Training
DCGAN (WGAN-GP, $n_c=5$)	221.51	$\sim$2 h
$\hookrightarrow$ corrected (std, $n_c=1$)	98.99	$\sim$3 h
VQ-VAE+PixelCNN	187.54	$\sim$4 h
DDPM+EMA	**24.36**	$\sim$12 h

Not compute-normalised; shows best feasible config per model. All single-run. Original DCGAN was under-trained (see §DCGAN Fix).

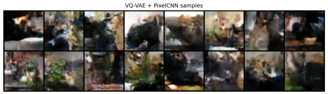
Why such a large gap?

- DDPM has a well-defined MSE objective: smooth, stable gradients
- EMA weights reduce FID by 55% vs. non-EMA at 50 steps
- GAN mode collapse limits diversity despite WGAN-GP
- VQ-VAE: good reconstruction, PixelCNN prior limits sample quality

Consistent with *Diffusion Models Beat GANs* (Dhariwal & Nichol, 2021)

Generated Samples: Side by Side

(a) DCGAN
FID 221.51

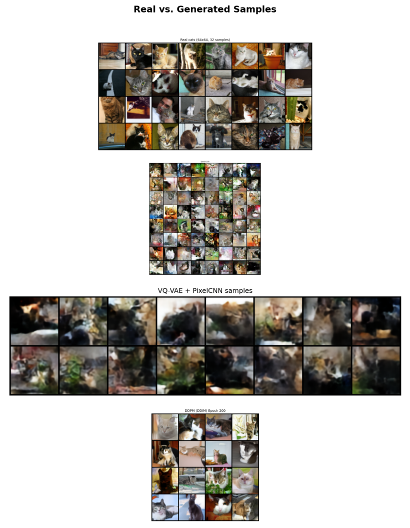


(b) VQ-VAE+PixelCNN
FID 187.54

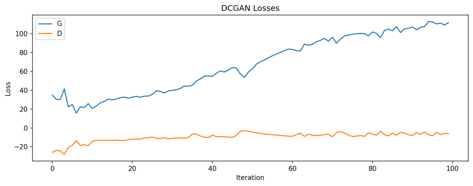
(c) **DDPM**
FID 24.36

DCGAN: artifacts, limited diversity | VQ-VAE: plausible but blurry | **DDPM: sharpest, most coherent**

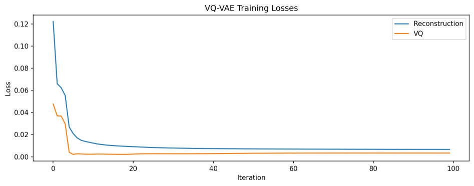
Real vs. Generated



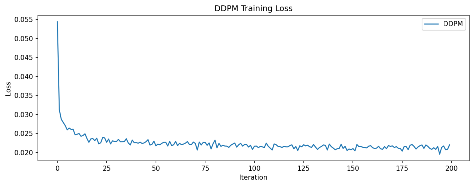
Training Dynamics



DCGAN (G vs. D) — adversarial oscillation



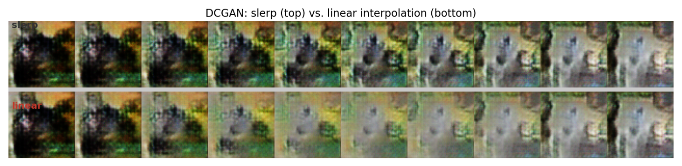
VQ-VAE (recon + VQ) — steady decrease



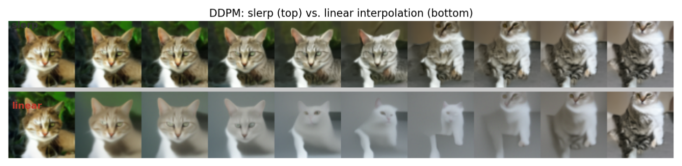
DDPM (ℓ_2 noise) — smooth convergence, most stable

DDPM's MSE objective gives the most stable training. DCGAN shows adversarial oscillation; VQ-VAE converges reliably with EMA codebook.

Latent Space Interpolation: Slerp vs. Linear



DCGAN — **slerp** (top) vs. linear (bottom)



DDPM — **slerp** (top) vs. linear (bottom)

Both methods shown. Linear interpolation (bottom rows) crosses a low-density region near the origin of $\mathcal{N}(0, I)$, producing blurrier intermediate frames — most visible in DDPM (washed-out midpoints). Slerp preserves the norm of the latent vector and yields smoother transitions.

VQ-VAE Reconstruction Quality

Top: input | Bottom: VQ-VAE reconstruction (epoch 100)
MSE = 0.0066 | VQ loss = 0.0033 | perplexity 403/512 (79%)

DCGAN Training Progression

From blurry noise to recognisable cats — but likely limited by low generator update budget (3 100 G-steps at n_c=5).

Experimental Grid: What We Tested

Model	Ablation / variant	Values tested	Metric
DCGAN	Latent dim z_{dim}	64, **128**, 256	FID, slerp smoothness
	Learning rate	1e-4, **2e-4**, 4e-4	FID
	Augmentation	off (**base**), on	FID
	Training data	cats only, **cats+dogs**	FID vs cats/dogs
DDPM	DDIM steps	10, 25, **50**, 100, 200	FID, wall time
	EMA weights	off, **on** (decay 0.9999)	FID
VQ-VAE	Codebook size K	1024 (collapsed) → **512**	perplexity, FID
Post-hoc fix	Standard DCGAN ($n_c = 1$)	150 ep, $g_{\text{ch}} = 128$	FID
	WGAN-GP $n_c = 2$	200 ep	FID

Bold = chosen baseline. Primary metric: FID on 5 000 generated vs 5 000 real images. Secondary: latent interpolation smoothness, reconstruction MSE (VQ-VAE).

DDPM: DDIM Steps and EMA Effect

FID vs. DDIM steps (non-EMA)

Steps	FID ↓	Speedup
10	66.92	100×
25	60.02	40×
50	**54.51**	**20×**
100	48.70	10×
200	39.81	5×

EMA at 50 steps: −55% FID

Weights	FID ↓
Without EMA	54.51
With EMA	**24.36**

50 steps: practical sweet spot (20× speedup).
EMA more impactful than doubling the steps.
Below 25 steps: FID degrades by 37% vs. 100 steps.

DCGAN: Latent Dimension and Learning Rate

z_{dim} ablation (3 seeds)

z_{dim}	FID mean$\pm$std	Δ
64	246.65 ± 5.57	-17.60
128 (baseline)	264.25 ± 20.21	-
256	266.05 ± 2.52	$+1.80$

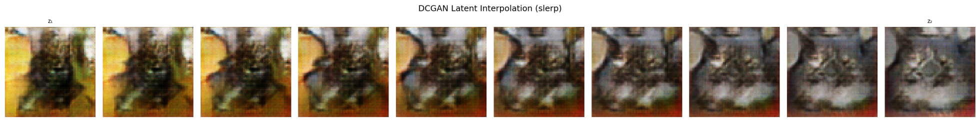
z_{64} best, but within baseline σ. High baseline variance ($\pm$20) shows seed sensitivity.

Learning rate ablation (3 seeds)

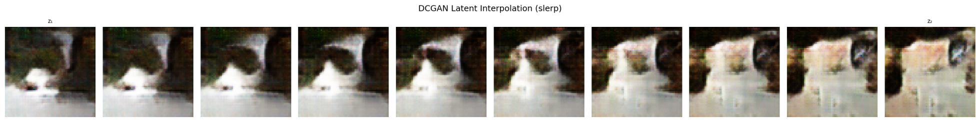
LR	FID mean$\pm$std	Δ
1×10^{-4}	295.11 ± 18.27	$+30.86$
2×10^{-4} (base)	264.25 ± 20.21	-
4×10^{-4}	$\mathbf{220.05 \pm 4.47}$	-44.20

LR has far more impact than z_{dim}.
Low LR: generator cannot keep up with the critic.

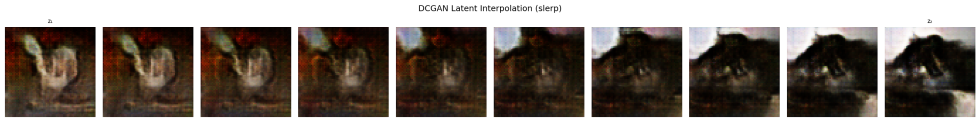
Latent Space Smoothness vs. z_{dim}



$z_{\text{dim}} = 64$



$z_{\text{dim}} = 128$ (baseline)



$z_{\text{dim}} = 256$

Slerp, 10 frames. Transitions smooth for all variants. z_{64} slightly softer; z_{128} and z_{256} indistinguishable.

Does Training on Dogs Help Cats?

Setup

- 12 500 cats + 12 500 dogs (Kaggle)
- Same DCGAN, 100 epochs, unconditional
- FID measured separately vs. cats and dogs

Results

Model	vs. Cats	vs. Dogs	vs. Combined
Cats only	221.51	-	-
Cats+Dogs	206.57	197.79	**190.39**

Combined FID (2 500 cats + 2 500 dogs) is more appropriate for a mixed-class model. 15-pt cat improvement is within seed variance (± 20) — suggestive, not conclusive.

Qualitative: samples show a mix — some clearly cat-like (round face, pointed ears), some dog-like (longer snout), many ambiguous dark quadruped silhouettes. No collapse to one class; model captures shared fur/eye features without cleanly

Cats+Dogs DCGAN (epoch 100)

DCGAN: Why FID Was So High and How It Was Fixed

Post-hoc diagnosis: generator update budget

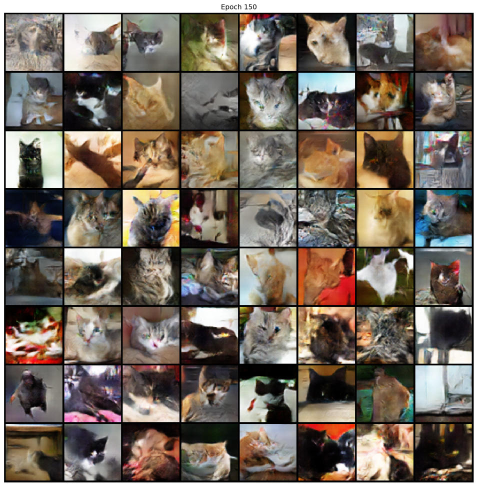
- $n_{\text{critic}} = 5$: 5 D-updates per G-update
- 156 batches/epoch $\Rightarrow$ only **3 100 G-steps** in 100 ep.
- Standard DCGAN ($n_c = 1$): **15 600** at same cost

Two corrected configs

- **Standard DCGAN**: switch to BCE loss; $n_c = 1$ is the correct standard (not a compromise), $g_{\text{ch}} = 128$, 150 ep. $\Rightarrow$ **23 400 G-steps**
- **WGAN-GP** $n_c = 2$: pragmatic fix; theoretically $n_c \geq 5$ required to preserve Lipschitz guarantee — proper fix ≈500 ep., 200 ep. $\Rightarrow$ **15 600 G-steps**

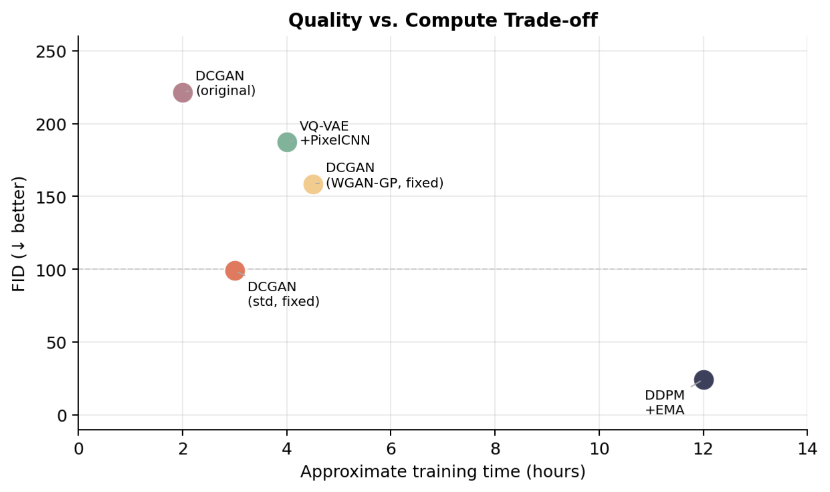
FID results

Config	G-steps	FID
Original	3 100	221.51
Std. DCGAN	23 400	**98.99**
WGAN-GP $n_c = 2$	15 600	158.48



Std. DCGAN (epoch 150)

Quality vs. Compute Trade-off



DDPM dominates in quality; DCGAN fastest. Fixed DCGAN closes the gap significantly.

Research Questions: Answers

RQ1: Architecture trade-off under 6 GiB VRAM

DDPM: best quality (FID 24.36, 12 h, 5.5 GiB). DCGAN: fastest (2 h, <2 GiB) but mode collapse. VQ-VAE: stable training, PixelCNN prior is the bottleneck.

RQ2: Does adding dogs improve cat generation?

Possibly, but not conclusively. Cat FID improves from 221.51 to 206.57, but the 15-pt gap is within the observed DCGAN seed variance (± 20). Class coverage and absence of hybrid artefacts would require a classifier-based analysis.

RQ3: Effect of latent dimension $z_{\mathbf{dim}} \in \{64, 128, 256\}$

Differences are within baseline seed variance (± 20); no firm conclusion about optimal z_{dim} can be drawn. $z = 64$ best on mean FID but not conclusively.

RQ4: DDIM speed-quality trade-off

50 steps: $20\times$ speedup, FID 54.51 (non-EMA) or 24.36 (with EMA). Below 25 steps quality degrades sharply. 50 steps is the practical sweet spot.

Conclusion

Summary

Model	FID	Strength
DDPM	**24.36**	quality
VQ-VAE	187.54	stability
DCGAN	221.51	speed

- Diffusion clearly outperforms GAN and VAE
- For DDPM, EMA more impactful than doubling DDIM steps
- DDIM enables practical use at 50 steps
- Dog images: suggestive improvement, not conclusive
- DCGAN bottleneck fixed: $7\times$ more G-updates

Main challenges

- 6 GiB VRAM: DDPM batch size limited to 16
- DCGAN mode collapse: WGAN-GP (label smoothing inactive under GP path)
- Codebook collapse at $K = 1024$: fixed with EMA, $K = 512$
- 8 h inference: solved with DDIM (50 steps, 25 min)
- DCGAN $n_{\text{critic}} = 5$: generator starvation fixed

Future work

- DDPM at 128×128 with gradient checkpointing
- Conditional DCGAN on cats+dogs
- Transformer prior (VQ-VAE-2 style)
- Cosine noise schedule

Thank you.

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